DEPLOYED SYSTEM PORTFOLIO

Stateful Multi-Agent B2B Lead Accelerator

A serverless production-grade outbound platform deployed on Google Cloud Run. Orchestrates decoupled React frontends and LangGraph pipelines to automate personalized lead mapping and adversarial stress-testing.

LangGraph · Next.js App Router · FastAPI Gateway · GCP Cloud Run

Project Scope & Demo Context

Confidentiality Notice

Because the production stack handles enterprise data pipelines, internal CRM APIs, and secure VPC networks, this repository serves as a **lightweight, self-contained developer demo**. It isolates the core agent loop and FastAPI logic, enabling localized evaluation while mirroring the production system's complete mechanics.

In production, manual lead research and cold copy-compilation are major bottlenecks. I designed this stateful architecture to automate account mapping, compile grounded outbound hooks, run adversarial prospect objection loops, and coach copy quality prior to pipeline sync.

Deployed Production Architecture

The enterprise deployment runs on **Google Cloud Run** using containerized microservices isolated behind private secure VPC subnets to manage scale and zero idle costs.

Next.js App
SDR Dashboard UI

FastAPI Gateway
LangGraph Engine

CrewAI Container
Research Partner

PostgreSQL SQL
Cloud SQL Checkpoints

The Stateful Outbound Lifecycle

The Campaign pipeline flows asynchronously through 5 key stages, utilizing database checkpointers to ensure failover recovery.

01

Researcher

Scans ICP parameters and structures initial prospect metadata.

02

Approval Gate

Halts graph and serializes session state to Cloud SQL.

03

Personalizer

Compiles hooks utilizing grounded case studies.

04

Objection Sim

Runs adversarial roleplay simulating client objections.

05

CRM Coach

Evaluates responses, routing back if score fails threshold.

High-Throughput Lead Verification & Ingestion Pipeline

To support enterprise-grade demand generation, the platform scales to process and verify **millions of contacts annually**, isolating tasks into high-throughput asynchronous execution branches and implementing multi-LLM data quality checks.

Distributed Worker Queues

Bypasses HTTP gateway timeouts during deep contact enrichment cycles by decoupling requests through a **GCP Pub/Sub task queue**. Scales background container workers dynamically to process high-volume lists in parallel.

Multi-LLM AI Consensus

Validates lead information using a multi-LLM consensus validation loop. Runs parallel extractions across **Claude 3.5** (precision extraction), **GPT-4o** (objection vetting), and **Gemini 1.5 Pro** (deep-context profiling).

Idempotent CRM Sync

Features a transactional checkpointer (PostgreSQL Cloud SQL) ensuring database state rollback safety. Guarantees 100% idempotent updates to CRM systems, completely preventing duplicate record spikes or API rate limit lockouts.

Outbound Agent Core Capability Matrix

The system equips containerized microservice agents with specialized tooling and analytical cognitive profiles:

Lead Finder

- Web Search (Google, Tavily)
- LinkedIn ICP Scrapers
- Contact Verification

Personalizer

- Case Study Synthesis
- Vector Playbook Grounding
- Multi-Hook Synthesis

Objection Sim

- adversarial critic Roleplay
- Friction Modeling
- Skeptical Persona Generator

CRM Coach

- DeepEval G-Eval Grades
- Feedback Loop Synthesizer
- State checkpointer Writer

Enterprise Engineering Hurdles Resolved

Asynchronous Scaling

TIMEOUTS

Intensive multi-agent research loops take between 30 to 90 seconds. Synchronous REST APIs resulted in connection timeouts between gateway and research containers.

Resolution

Decoupled executions using a **GCP Pub/Sub task queue**. The Personalizer posts tasks, transitions to `AWAITING_RESEARCH`, and releases the thread. Background worker containers complete research and trigger gateway resume webhooks.

Bulletproof Recovery

FAULT TOLERANCE

Serverless Cloud Run instances scale down dynamically. Campaigns risked data loss and redundant API expenses if active containers crashed mid-campaign.

Resolution

Configured an enterprise **PostgreSQL Cloud SQL checkpointer**. Since every node transition is committed as a strict database transaction, active containers instantly resume right where they left off without repeating expensive research.

Real-Time Guardrails

QUALITY CHECKS

Generative models are susceptible to hallucinating value metrics or numbers not explicitly supported by target account playbooks and source material.

Resolution

Built a **validation filter** directly inside the personalizer node. Powered by **DeepEval**, it checks generated pitches for **Groundedness**. If scores fall below 0.85, the pitch is rejected and automatically routed to self-correction.

Roadmap: Future Production Integrations

To further scale our B2B agent capabilities, the production deployment is currently integrating two advanced features:

Dense Retrieval RAG (Dense RAG)

Transitioning our standard local file context mapping into a highly structured **Dense RAG** pipeline. By indexing all case studies, company playbooks, and competitor intelligence into a vector database, our SDR Personalizer executes semantic search queries to extract exact passages. This completely prevents hallucinations by ensuring every outbound hook is directly backed by proprietary corporate text.

Autonomous Live Web Search Discovery

Integrating dynamic web search capabilities inside our Lead Finder. Outfitting researchers with Google Search, Tavily, and LinkedIn API pipelines enables them to search active business directories dynamically. This allows the system to autonomously discover new leads, pull real-time industry news, and trigger outreach loops without requiring static manual target files.

High-Scalability Enterprise Stack Architecture (8M+ Contacts/Year)

To scale this B2B Lead Accelerator to an enterprise-grade high-throughput environment, the system design transitions from single-process operations into a highly distributed, parallel microservice topology coordinated by GCP messaging channels and replicated PostgreSQL databases:

The Five Scalable Infrastructure Layers

1. Enterprise Ingestion & Verification Layer: Employs horizontally scaled **GCP Cloud Run Jobs** using Token Bucket rate-limiting and rotating proxy pools to verify MX records and scrape LinkedIn data for 22,000+ prospects daily.

2. Asynchronous Event Bus: Decouples long-running multi-agent research operations out-of-band using a highly resilient **GCP Pub/Sub** queue, featuring a Dead Letter Queue (DLQ) for self-healing error isolation.

3. Scalable Compute & State Orchestration: Coordinated by **LangGraph Orchestrator Worker Pools**, scaling database checkpoints horizontally via a replicated **GCP Cloud SQL (PostgreSQL)** setup with **PgBouncer** connection poolers.

4. Decoupled Agent-to-Agent Microservices: Separates execution concerns by hosting the CrewAI Research Partner and Objection Simulator in standalone auto-scaling containers, utilizing local fine-tuned **Llama-3-8B** GPU models via Vertex AI to slash API token expenses.

5. Transactional Delivery & CRM Synchronization: Enforces a strict **Transactional Outbox Pattern** (`crm_outbox` ledger commits) inside database transaction blocks, ensuring 100% idempotent updates to HubSpot/Salesforce CRMs with exponential backoff retries.

💡 AI-First Product Leadership Value

This architecture shows a robust understanding of production AI patterns: state graph predictability (vs. fragile prompt chains), serverless decoupled microservices scaling to zero, and structured database-backed approval gates ensuring brand safety.

- ✓ State Graph Reliability
- ✓ Decoupled Compute Savings
- ✓ Human-in-the-Loop Safeguards